Spatio-temporal Analysis of Aerosol or Air Quality Indicators using Remote Sensing and GIS

Project Report Submitted in fulfillment of the Requirements for the Award of the Internship of Summer Training Program Space Science Technology

Remote Sensing, GIS, Artificial Intelligence, and Python

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1. Title

Spatio-temporal Analysis of Aerosol or Air Quality Indicators using Remote Sensing and GIS.

2. Objective

To analyze the spatial and temporal distribution of aerosol concentration or air quality indicators using satellite data, and to identify patterns related to seasonal variation, urbanization, and industrial activity.

3. Study Area

Kolkata Metropolitan Area, West Bengal, India

The study was conducted over the Kolkata Metropolitan Area (KMA) in the state of West Bengal, India. Kolkata is one of the largest metropolitan cities in eastern India and is characterised by high population density, extensive transportation networks, industrial activities, and rapid urban development. These factors contribute significantly to aerosol emissions and deteriorating air quality, making the region highly suitable for satellite-based air quality analysis.

The Area of Interest (AOI) was manually delineated using the Geometry tools available in the Google Earth Engine Code Editor to encompass the urban extent of Kolkata and its surrounding regions. The approximate geographic extent of the selected area lies between 22.45°N–22.75°N latitude and 88.20°E–88.55°E longitude.

Kolkata experiences a tropical wet-and-dry climate, with hot and humid summers, a monsoon season, and mild winters. Meteorological conditions such as humidity, wind patterns, rainfall, and temperature strongly influence the concentration and dispersion of atmospheric aerosols throughout the year. These seasonal variations make Kolkata an appropriate study area for analysing the spatial distribution of aerosol loading.

An important geographical feature within the study area is the Hooghly River, which flows through the city and influences local atmospheric conditions. The selected AOI includes urban, industrial, residential, vegetated, and water-covered regions, enabling comparison of aerosol concentrations across different land-use types. This diversity makes the study area well suited for identifying pollution hotspots and analysing spatial patterns of air quality using satellite-derived Aerosol Optical Depth (AOD) data.

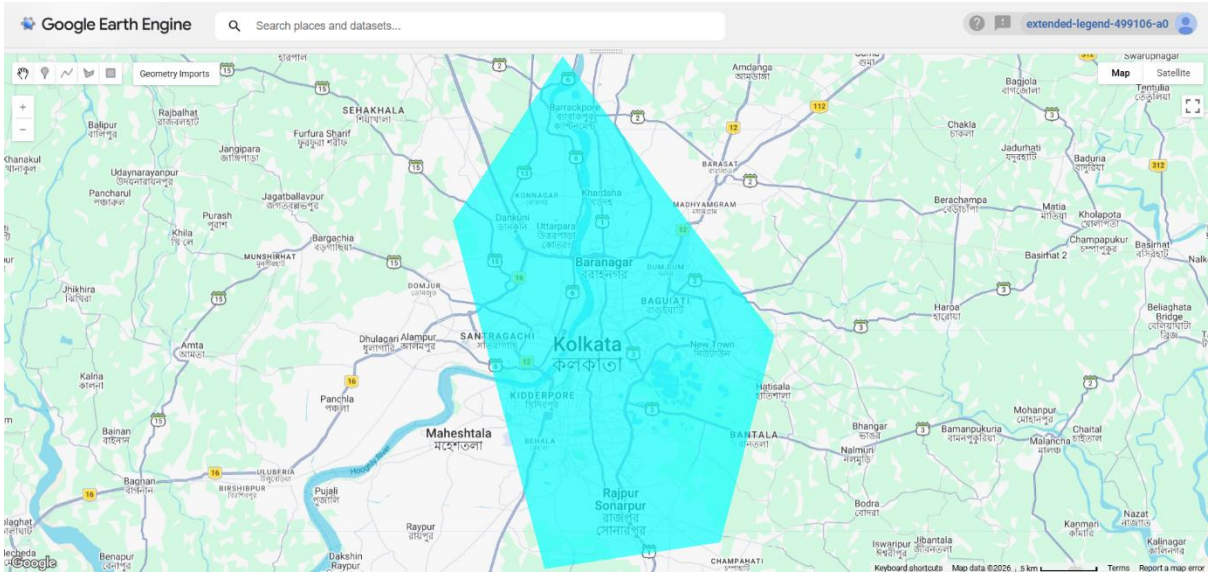


Figure 1. Area of Interest (AOI): Kolkata Metropolitan Area.

4. Data Used

Satellite Data:

- MODIS MCD19A2 Version 6.1 Aerosol Optical Depth (AOD) Dataset
  Time Period: November 2023
  Spatial Resolution: 1 km

Band Used:

• Optical_Depth_047

Ancillary Data:

• Area of Interest (AOI) created manually in Google Earth Engine using the Geometry tool.

Software Used:

• Google Earth Engine

5. Methodology

Step 1: Area of Interest (AOI) Selection
The Kolkata Metropolitan Area was selected as the study area due to its high population density, rapid urbanization, industrial activities, and recurring air quality concerns. The Area of Interest (AOI) representing the study region was manually delineated using the Geometry tools available in the Google Earth Engine Code Editor.

Step 2: MODIS Aerosol Optical Depth (AOD) Data Acquisition and Preparation

MODIS MCD19A2 Version 6.1 Aerosol Optical Depth (AOD) imagery for November 2023 was obtained from the Google Earth Engine Data Catalog. The Optical_Depth_047 band was selected for analysis. The image collection was filtered using the selected Area of Interest (AOI) and the specified time period. The mean AOD image was generated by averaging all available observations during the selected month and clipping the output to the AOI.

Step 3: Mean Aerosol Optical Depth (AOD) Mapping

The processed mean AOD image was visualised using an appropriate colour gradient ranging from green (low aerosol concentration) to red (high aerosol concentration). This thematic map was used to examine the spatial distribution of atmospheric aerosols across the Kolkata Metropolitan Area.

Step 4: Pollution Hotspot Identification

Pollution hotspot zones were identified by applying an Aerosol Optical Depth threshold to the mean AOD image. Pixels having AOD values greater than the selected threshold were classified as pollution hotspots and visualised as red regions on the map.

Step 5: Statistical Analysis

Statistical analysis was performed using Google Earth Engine to quantify the aerosol characteristics within the study area. The minimum, maximum and mean Aerosol Optical Depth (AOD) values were computed from the processed raster dataset. In addition, the total area occupied by the identified pollution hotspots was calculated using pixel area analysis and expressed in square kilometres.

Workflow:-

AOI

↓

MODIS MCD19A2 AOD Data

↓

Mean AOD Map

↓

Pollution Hotspots

↓

Statistical Analysis

6. Results

6.1 Mean Aerosol Optical Depth (AOD) Distribution

The mean Aerosol Optical Depth (AOD) map illustrates the spatial distribution of atmospheric aerosol concentration across the Kolkata Metropolitan Area during November 2023. Aerosol Optical Depth is an important indicator of the quantity of suspended particulate matter present in the atmosphere. Lower AOD values indicate relatively cleaner atmospheric conditions, whereas higher values suggest increased aerosol loading.

The generated map reveals noticeable spatial variation in aerosol concentration throughout the study area. Most regions exhibit moderate AOD values, represented by yellow shades, while several localized patches display comparatively higher aerosol concentrations, shown in orange and red. Areas with relatively lower AOD values appear in green, indicating comparatively cleaner atmospheric conditions. These variations reflect differences in urban development, transportation activities, industrial emissions, and surrounding land cover.

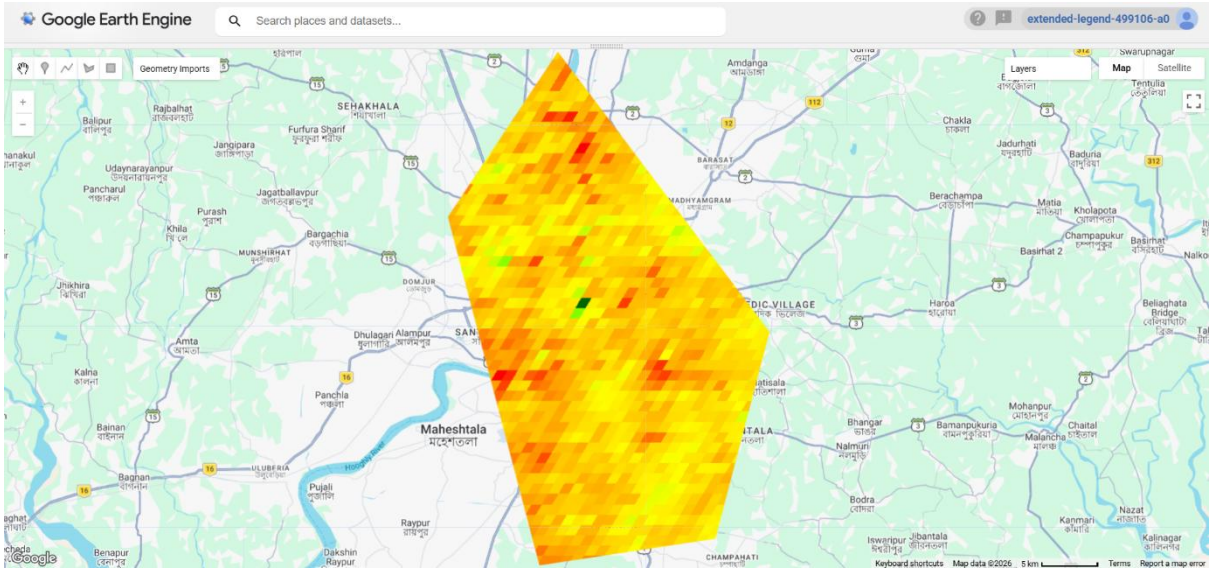


Figure 2. Mean Aerosol Optical Depth (AOD) distribution over the Kolkata Metropolitan Area for November 2023.

The computed AOD values ranged from **665.68** to **1130.25**, with an average value of **953.11** for the selected period.

The observed AOD distribution suggests that aerosol concentration is not uniform across Kolkata. Regions exhibiting comparatively higher AOD values are likely influenced by dense urbanization, heavy traffic, industrial activities, and construction-related emissions. In contrast, areas with lower AOD values are generally associated with vegetation, open spaces, or comparatively less urbanized surroundings. This spatial variability

demonstrates the usefulness of satellite-derived aerosol data for regional air quality assessment.

6.2 Air Pollution Hotspot Identification

Pollution hotspot zones were identified by applying a threshold of **1000** to the mean Aerosol Optical Depth (AOD) image. Pixels exhibiting comparatively higher AOD values were classified as pollution hotspots. These regions indicate relatively elevated atmospheric aerosol concentrations and are likely influenced by dense urban development, vehicular emissions, industrial activities, and other anthropogenic sources. The hotspot map provides a clear spatial representation of areas experiencing higher levels of atmospheric pollution within the study area.

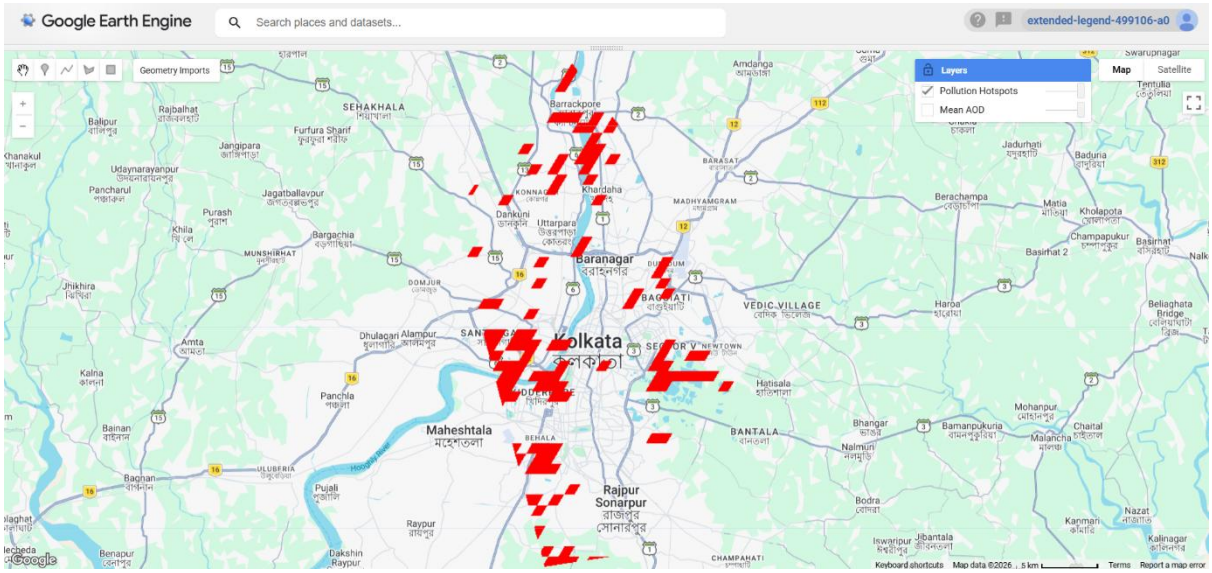


Figure 3. Pollution hotspot zones identified from the mean Aerosol Optical Depth (AOD) map over Kolkata Metropolitan Area.

The estimated total area occupied by pollution hotspots within the study area was approximately **91.95 km²**.

6.3 Statistical Analysis of AOD

Statistical analysis of the processed Aerosol Optical Depth (AOD) data was carried out using Google Earth Engine to summarize the overall aerosol conditions within the study area. The minimum, maximum and mean AOD values were calculated from the processed raster, while the total area occupied by pollution hotspots was also estimated.

These statistics provide a quantitative assessment of the spatial distribution of aerosol concentration across the Kolkata Metropolitan Area.



Figure 4. Statistical summary of Aerosol Optical Depth (AOD) derived from Google Earth Engine.

Parameter	Value
Minimum AOD	665.68
Maximum AOD	1130.25
Mean AOD	953.11
Pollution Hotspot Area	91.95 km²

The relatively high mean AOD value indicates a noticeable aerosol burden over the study area during the selected period. The calculated hotspot area of approximately 91.95 km² suggests that a substantial portion of the

metropolitan region experienced comparatively elevated aerosol concentrations, which are generally associated with dense urban development, industrial activities and transportation corridors.

7. Discussion

The spatial distribution of Aerosol Optical Depth (AOD) over the Kolkata Metropolitan Area revealed noticeable variations in aerosol concentration across the study area. The mean AOD map indicated that most urban regions experienced moderate to high aerosol loading, while relatively lower values were observed over areas with greater vegetation cover and fewer anthropogenic activities. These variations demonstrate the influence of land use and human activities on regional air quality.

The identified pollution hotspots were primarily concentrated over densely populated and industrial parts of the metropolitan area. These regions are characterised by heavy vehicular traffic, industrial emissions, construction activities, and commercial development, all of which contribute significantly to atmospheric aerosol concentrations. In contrast, areas containing open spaces, water bodies, and vegetation generally exhibited comparatively lower aerosol levels due to reduced emission sources and improved natural ventilation.

The statistical analysis further supported these observations. The mean AOD value indicated an overall moderate to high aerosol burden during the selected period, while the estimated hotspot area of approximately 91.95 km² highlighted the spatial extent of regions experiencing relatively elevated aerosol concentrations. Such quantitative information is valuable for identifying priority areas requiring pollution monitoring and mitigation.

Overall, the results demonstrate that satellite-derived AOD data provide an effective means of monitoring regional air quality and identifying pollution-prone areas. Although Aerosol Optical Depth represents the total atmospheric aerosol content rather than ground-level pollutant concentration, it serves as a reliable indicator of atmospheric pollution patterns and enables large-scale environmental assessment where dense ground monitoring networks may not be available.

8. Limitations

Although the adopted methodology provides an effective assessment of aerosol distribution over the study area, certain inherent limitations should be considered while interpreting the results:

- The analysis was based on MODIS Aerosol Optical Depth (AOD) data for November 2023 and therefore represents air quality conditions during the selected period only.

- Aerosol Optical Depth reflects the total atmospheric aerosol loading and does not directly measure ground-level pollutant concentrations.

- The spatial resolution of the MODIS dataset may not capture small-scale variations in pollution within densely built urban environments.

- The results were not validated using ground-based air quality observations such as CPCB monitoring station data.

9. Future Scope of the Study

The present study demonstrates the effectiveness of satellite-based aerosol monitoring for urban air quality assessment. Future studies may incorporate multi-year satellite observations to evaluate long-term trends in aerosol concentration and assess the impact of seasonal variations on air quality. The integration of Sentinel-5P atmospheric products, such as nitrogen dioxide (NO_2), sulphur dioxide (SO_2), and carbon monoxide (CO), would provide a more comprehensive understanding of urban pollution.

Future work may also include the use of ground-based observations from the Central Pollution Control Board (CPCB) for validation of satellite-derived results, thereby improving the reliability of the analysis. In addition, advanced GIS techniques and machine learning models can be employed to predict future pollution hotspots and identify the influence of urban growth, traffic density, and industrial expansion on air quality. Such studies can contribute significantly to sustainable urban planning, environmental monitoring and public health management.

10. Conclusion

This study successfully analysed the spatial distribution of Aerosol Optical Depth (AOD) over the Kolkata Metropolitan Area using MODIS satellite data and Google Earth Engine. The generated AOD map provided a clear representation of aerosol concentration across the study area, while hotspot analysis identified regions experiencing relatively high levels of atmospheric aerosols. Statistical analysis further quantified the spatial characteristics of aerosol distribution by determining the minimum, maximum, and mean AOD values, along with the total extent of identified pollution hotspots.

The results indicate that higher aerosol concentrations are generally associated with densely urbanised and industrial regions, reflecting the influence of anthropogenic activities such as transportation, industrial emissions, and urban development. Conversely, comparatively lower aerosol concentrations were observed over areas containing vegetation and open spaces, highlighting the role of natural land cover in improving environmental conditions.

Overall, this project demonstrates that satellite remote sensing integrated with GIS provides an efficient and reliable approach for monitoring regional air quality over large areas. The methodology enables rapid identification of pollution-prone zones and supports evidence-based environmental assessment. Such geospatial analyses can assist researchers, urban planners, and policymakers in developing effective pollution control strategies and promoting sustainable urban development.

11. References

- Google Earth Engine, https://earthengine.google.com/
- Google Earth Engine Data Catalog, https://developers.google.com/earth-engine/datasets
- NASA - MODIS (Moderate Resolution Imaging Spectroradiometer) Mission

12. Appendix: Google Earth Engine Script

The complete Google Earth Engine (GEE) script used for Aerosol Optical Depth (AOD) analysis, pollution hotspot identification, and statistical computation is provided below. The script includes data acquisition, preprocessing, visualization, hotspot extraction, and area calculation performed for the Kolkata Metropolitan Area.

```
var geometry =
    /* color: #00ffff */
    /* shown: false */
    ee.Geometry.Polygon(
        [[[88.34123960929273, 22.814853553694935],
          [88.24510923819898, 22.681874719573557],
          [88.32513843427773, 22.400671122440492],
          [88.47894702802773, 22.422253744984182],
          [88.5263255680668, 22.589088168094047]]]);

var aoi = geometry;
Map.centerObject(aoi, 11);

var aod = ee.ImageCollection('MODIS/061/MCD19A2_GRANULES')
```

```
    .filterBounds(aoi)
    .filterDate('2023-11-01', '2023-11-30')
    .select('Optical_Depth_047');

print('AOD Collection', aod);

var meanAOD = aod.mean().clip(aoi);

print('Mean AOD', meanAOD);

Map.addLayer(
  meanAOD,
  {
    min: 700,
    max: 1100,
    palette: [
      '006400',
      '7FFF00',
      'FFFF00',
      'FFA500',
      'FF0000'
    ]
  },
  'Mean AOD'
);

var stats = meanAOD.reduceRegion({
  reducer: ee.Reducer.minMax(),
  geometry: aoi,
  scale: 1000,
  maxPixels: 1e13
});

print('AOD Min Max', stats);

print(
  'Mean AOD Value',
  meanAOD.reduceRegion({
    reducer: ee.Reducer.mean(),
    geometry: aoi,
    scale: 1000,
    maxPixels: 1e13
  })
);

var hotspot = meanAOD.gt(1000);

Map.addLayer(
  hotspot.updateMask(hotspot),
```

```
  {palette:['red']},
  'Pollution Hotspots'
);

var hotspotArea = hotspot.multiply(ee.Image.pixelArea());

var area = hotspotArea.reduceRegion({
  reducer: ee.Reducer.sum(),
  geometry: aoi,
  scale: 1000,
  maxPixels: 1e13
});

print('Hotspot Area (sq m)', area);

print('Hotspot Area (sq km)', ee.Number(area.get('Optical_Depth_047')).divide(1e6));
```